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Inventor(s)	Lee; GunHyuck et al.

Semiconductor device and method of disposing electrical components over side surfaces of interconnect substrate

Abstract

A semiconductor device has an interconnect substrate with a conductive via. A first electrical component is disposed over a major surface of the interconnect substrate. An electrical interconnect compound is disposed over the conductive via exposed from a side surface of the interconnect substrate. The electrical interconnect compound can be applied with a tilt nozzle oriented at an angle. A second electrical component is disposed on the electrical interconnect compound on the conductive via exposed from the side surface of the interconnect substrate. A plurality of second electrical components can be disposed on two or more side surfaces of the interconnect substrate. The interconnect substrate can have a plurality of stacked conductive vias and the second electrical component is disposed over the stacked conductive vias. An encapsulant is deposited over the first electrical component and interconnect substrate. A shielding layer can be formed over the encapsulant.

Inventors: **Lee; GunHyuck (Incheon, KR), Lee; HeeSoo (Incheon, KR), Son; SangHyun (Seoul, KR), Hwang; Bokyeong (Incheon, KR)**

Applicant: **STATS ChipPAC Pte. Ltd.** (Singapore, SG)

Family ID: **1000008781137**

Assignee: **STATS ChipPAC Pte. Ltd. (Singapore, SG)**

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Primary Examiner: Han; Jonathan

Attorney, Agent or Firm: PATENT LAW GROUP: Atkins and Associates, P.C.

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates in general to semiconductor devices and, more particularly, to a semiconductor device and method of disposing a plurality of electrical components over one or more side surfaces of an interconnect substrate.

BACKGROUND OF THE INVENTION

(2) Semiconductor devices are commonly found in modern electronic products. Semiconductor devices perform a wide range of functions, such as signal processing, high-speed calculations, transmitting and receiving electromagnetic signals, controlling electronic devices, photo-electric, and creating visual images for television displays. Semiconductor devices are found in the fields of

communications, power conversion, networks, computers, entertainment, and consumer products. Semiconductor devices are also found in military applications, aviation, automotive, industrial controllers, and office equipment.
(3) Semiconductor devices, particularly in high frequency applications, such as radio frequency (RF) wireless communications, often contain one or more integrated passive devices (IPDs) to perform necessary electrical functions. Multiple semiconductor die, IPDs, and other electrical components can be integrated into a system in package (SIP) module or other wafer level package (WLP) for higher density in a small space and extended electrical functionality. Within the SIP module, the electrical components are mounted to a substrate for structural support and electrical interconnect. An encapsulant is deposited over the electrical components and substrate.
(4) Most if not all SIP modules and WLP arrange the electrical components on a major surface of the substrate. As the SIP module functionality continues to expand, more and more electrical components are placed on the major surface of the substrate and the SIP module becomes larger in surface area. Yet the preferred trend should be to make the SIP module smaller while increasing functionality.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1***a*-**1***c* illustrate a semiconductor wafer with a plurality of semiconductor die separated by a saw street;
(2) FIGS. **2***a*-**2***h* illustrate a process of disposing electrical components over one or more side surfaces of an interconnect substrate to form an SIP module;
(3) FIGS. **3***a*-**3***h* illustrate further detail of disposing electrical components over one or more side surfaces of the interconnect substrate;
(4) FIGS. **4***a*-**4***g* illustrate the use of carriers to dispose electrical components over a side surface of the interconnect substrate;
(5) FIGS. **5***a*-**5***b* illustrate disposing electrical components over multiple side surfaces of the interconnect substrate;
(6) FIG. **6** illustrates a shielding layer over the SIP module with electrical components disposed over one or more side surfaces of the interconnect substrate; and
(7) FIG. **7** illustrates a printed circuit board (PCB) with different types of packages mounted to a surface of the PCB.
DETAILED DESCRIPTION OF THE DRAWINGS
(8) The present invention is described in one or more embodiments in the following description with reference to the figures, in which like numerals represent the same or similar elements. While the invention is described in terms of the best mode for achieving the invention's objectives, it will be appreciated by those skilled in the art that it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the appended claims and their equivalents as supported by the following disclosure and drawings. The term “semiconductor die” as used herein refers to both the singular and plural form of the words, and accordingly, can refer to both a single semiconductor device and multiple semiconductor devices.
(9) Semiconductor devices are generally manufactured using two complex manufacturing processes: front-end manufacturing and back-end manufacturing. Front-end manufacturing involves the formation of a plurality of die on the surface of a semiconductor wafer. Each die on the wafer contains active and passive electrical components, which are electrically connected to form functional electrical circuits. Active electrical components, such as transistors and diodes, have the ability to control the flow of electrical current. Passive electrical components, such as

capacitors, inductors, and resistors, create a relationship between voltage and current necessary to perform electrical circuit functions.
(10) Back-end manufacturing refers to cutting or singulating the finished wafer into the individual semiconductor die and packaging the semiconductor die for structural support, electrical interconnect, and environmental isolation. To singulate the semiconductor die, the wafer is scored and broken along non-functional regions of the wafer called saw streets or scribes. The wafer is singulated using a laser cutting tool or saw blade. After singulation, the individual semiconductor die are mounted to a package substrate that includes pins or contact pads for interconnection with other system components. Contact pads formed over the semiconductor die are then connected to contact pads within the package. The electrical connections can be made with conductive layers, bumps, stud bumps, conductive paste, or wirebonds. An encapsulant or other molding material is deposited over the package to provide physical support and electrical isolation. The finished package is then inserted into an electrical system and the functionality of the semiconductor device is made available to the other system components.
(11) FIG. **1***a* shows a semiconductor wafer **100** with a base substrate material **102**, such as silicon, germanium, aluminum phosphide, aluminum arsenide, gallium arsenide, gallium nitride, indium phosphide, silicon carbide, or other bulk material for structural support. A plurality of semiconductor die or components **104** is formed on wafer **100** separated by a non-active, inter-die wafer area or saw street **106**. Saw street **106** provides cutting areas to singulate semiconductor wafer **100** into individual semiconductor die **104**. In one embodiment, semiconductor wafer **100** has a width or diameter of 100-450 millimeters (mm).
(12) FIG. **1**B shows a cross-sectional view of a portion of semiconductor wafer **100**. Each semiconductor die **104** has a back or non-active surface **108** and an active surface **110** containing analog or digital circuits implemented as active devices, passive devices, conductive layers, and dielectric layers formed within the die and electrically interconnected according to the electrical design and function of the die. For example, the circuit may include one or more transistors, diodes, and other circuit elements formed within active surface **110** to implement analog circuits or digital circuits, such as digital signal processor (DSP), application specific integrated circuits (ASIC), memory, or other signal processing circuit. Semiconductor die **104** may also contain IPDs, such as inductors, capacitors, and resistors, for RF signal processing.
(13) An electrically conductive layer **112** is formed over active surface **110** using PVD, CVD, electrolytic plating, electroless plating process, or other suitable metal deposition process. Conductive layer **112** can be one or more layers of aluminum (Al), copper (Cu), tin (Sn), nickel (Ni), gold (Au), silver (Ag), or other suitable electrically conductive material. Conductive layer **112** operates as contact pads electrically connected to the circuits on active surface **110**.
(14) An electrically conductive bump material is deposited over conductive layer **112** using an evaporation, electrolytic plating, electroless plating, ball drop, or screen printing process. The bump material can be Al, Sn, Ni, Au, Ag, Pb, Bi, Cu, solder, and combinations thereof, with an optional flux solution. For example, the bump material can be eutectic Sn/Pb, high-lead solder, or lead-free solder. The bump material is bonded to conductive layer **112** using a suitable attachment or bonding process. In one embodiment, the bump material is reflowed by heating the material above its melting point to form balls or bumps **114**. In one embodiment, bump **114** is formed over an under bump metallization (UBM) having a wetting layer, barrier layer, and adhesive layer. Bump **114** can also be compression bonded or thermocompression bonded to conductive layer **112**. Bump **114** represents one type of interconnect structure that can be formed over conductive layer **112**. The interconnect structure can also use bond wires, conductive paste, stud bump, micro bump, or other electrical interconnect.
(15) In FIG. **1***c*, semiconductor wafer **100** is singulated through saw street **106** using a saw blade or laser cutting tool **118** into individual semiconductor die **104**. The individual semiconductor die **104** can be inspected and electrically tested for identification of known good die or unit (KGD/KGU)

post singulation.

(16) FIGS. **2***a*-**2***h* illustrate a process of disposing electrical components over a side surface of an interconnect substrate to form an SIP module. FIG. **2***a* shows a cross-sectional view of multi-layered interconnect substrate **120** including conductive layers **122** and insulating layer **123**. Conductive layer **122** can be one or more layers of Al, Cu, Sn, Ni, Au, Ag, or other suitable electrically conductive material. Conductive layer **122** provides horizontal electrical interconnect across substrate **120** and vertical electrical interconnect vias **124** between top major surface **126** and bottom major surface **128** of substrate **120**. Portions of conductive layer **122** can be electrically common or electrically isolated depending on the design and function of semiconductor die **104** and other electrical components. Insulating layer **123** contains one or more layers of silicon dioxide (SiO2), silicon nitride (Si3N4), silicon oxynitride (SiON), tantalum pentoxide (Ta2O5), aluminum oxide (Al2O3), solder resist, polyimide, benzocyclobutene (BCB), polybenzoxazoles (PBO), and other material having similar insulating and structural properties. Insulating layer **123** provides isolation between conductive layers **122**.

(17) In FIG. **2***b*, a plurality of electrical components **130***a*-**130***c* is mounted to surface **126** of interconnect substrate **120** and electrically and mechanically connected to conductive layers **122**. Electrical components **130***a*-**130***c* are each positioned over die attach locations **129***a*-**129***c* of substrate **120**, respectively, using a pick and place operation. For example, electrical component **130***a* can be similar to semiconductor die **104** from FIG. **1***c*, with active surface **110** and bumps **114** oriented toward surface **126** of substrate **120**. Electrical components **130***b* and **130***c* can be similar to semiconductor die **104**, although possibly having a different form and function, with active surface **110** and bumps **114** oriented toward surface **126** of substrate **120**. Alternatively, electrical components **130***a*-**130***c* can include other semiconductor die, semiconductor packages, surface mount devices, RF component, discrete electrical devices, or IPDs, such as a resistor, capacitor, and inductor. FIG. **2***c* illustrates electrical components **130***a*-**130***c* electrically and mechanically connected to conductive layers **122** and vertical interconnect vias **124** of substrate **120**.

(18) In FIG. **2***d*, an encapsulant or molding compound **136** is deposited over and around electrical components **130***a*-**130***c* on substrate **120** using a paste printing, compressive molding, transfer molding, liquid encapsulant molding, vacuum lamination, spin coating, or other suitable applicator. Encapsulant **136** can be polymer composite material, such as epoxy resin with filler, epoxy acrylate with filler, or polymer with proper filler. Encapsulant **136** is non-conductive, provides structural support, and environmentally protects the semiconductor device from external elements and contaminants. Electrical components **130***a*-**130***c*, as mounted to interconnect substrate **120** and covered by encapsulant **136**, constitute SIP module or WLP **138**.

(19) In FIG. **2***e*, electrical components **140***a* and **140***b* are positioned over side surface **141** of interconnect substrate **120** using a pick and place operation. Side surface **141** is a surface of interconnect substrate **120** between and normal to major surface **126** and major surface **128**. Electrical components **140***a*-**140***b* can be semiconductor die (similar to semiconductor die **104**), semiconductor packages, surface mount devices, RF component, discrete electrical devices, or IPDs, such as a resistor, capacitor, and inductor. External connection terminal **142***a* and **142***b* of electrical components **140***a* and **140***b* are aligned with electrical interconnect compound **144**, such as solder or electrically conductive paste. Electrical interconnect compound **144** makes electrical connection between external connection terminals **142***a* and **142***b* and conductive layer **122** and vertical interconnect vias **124** of interconnect substrate **120**.

(20) FIG. **2***f* illustrates electrical components **140***a*-**140***b* electrically and mechanically connected to side surface **141** and conductive layers **122** and vertical interconnect vias **124** of interconnect substrate **120**. In the prior art, electrical components like **140***a* and **140***b* may have been placed over a major surface of the interconnect substrate, creating the need for a larger surface area to accommodate the components. Instead, electrical components **140***a* and **140***b* are disposed over side surface **141** of interconnect substrate **120** thereby reducing the surface area of surface **126** of

SIP module **138** dedicated to components.

(21) In another embodiment, continuing from FIG. **2***d*, electrical components **148***a* and **148***b* are positioned over side surface **141** of interconnect substrate **120** using a pick and place operation, as in FIG. **2***g*. For example, electrical components **148***a* and **148***b* can be similar to semiconductor die **104** from FIG. **1***c*, although having a different form and function, with contact pads **149***a* and **149***b* oriented toward side surface **141** of substrate **120**. Side surface **141** is a surface of interconnect substrate **120** between and normal to major surface **126** and major surface **128**. Alternatively, electrical components **148***a*-**148***b* can be other semiconductor die, semiconductor packages, surface mount devices, RF component, discrete electrical devices, or IPDs, such as a resistor, capacitor, and inductor. External connection terminal **149***a* and **149***b* of electrical components **148***a* and **148***b* are aligned with electrical interconnect compound **144**, such as solder or electrically conductive paste. Electrical interconnect compound **144** makes electrical connection between external connection terminals **142***a* and **142***b* and conductive layer **122** and vertical interconnect vias **124** of interconnect substrate **120**. Components having a similar function are assigned the same reference number in the figures.

(22) FIG. **2***h* illustrates electrical components **148***a*-**148***b* electrically and mechanically connected to side surface **141** and conductive layers **122** and vertical interconnect vias **124** of interconnect substrate **120**. In the prior art, electrical components like **148***a* and **148***b* may have been placed over a major surface of the interconnect substrate, creating the need for a larger surface area to accommodate the components. Instead, electrical components **148***a* and **148***b* are disposed over side surface **141** of interconnect substrate **120** thereby reducing the surface area of surface **126** of SIP module **138** dedicated to components.

(23) FIG. **3***a* illustrates SIP module **138** with encapsulant **136** deposited over electrical components **130***a*-**130***c* and interconnect substrate **120**. FIG. **3***b* is a side view of SIP module **138**, from FIG. **3***a*, showing insulating layers **123** and conductive layers **122** and vertical interconnect via **124** exposed from side surface **141**. In another embodiment, FIG. **3***c* is a side view of SIP module **138** showing insulating layers **123** and conductive layers **122** and stacked vertical interconnect vias **124** exposed from side surface **141**. In another embodiment, FIG. **3***d* is a side view of SIP module **138** showing insulating layers **123**, copper clad laminate (CCL) layer **150**, conductive layers **122**, and offset stacked vertical interconnect vias **124** exposed from side surface **141**.

(24) FIG. **3***e* illustrates SIP module **138** with encapsulant **136** deposited over electrical components **130***a*-**130***c* and interconnect substrate **120**. Electrical interconnect compound **144**, such as solder or electrically conductive paste, is deposited over the exposed vertical interconnect vias **124**. FIG. **3***f* is a side view of SIP module **138**, from FIG. **3***e*, showing insulating layers **123** and conductive layers **122** and electrical interconnect compound **144** deposited over the exposed vertical interconnect vias **124** on side surface **141**. Electrical interconnect compound **144** can be positioned horizontally or vertically and electrical components **140***a*-**140***b* are aligned to the electrical interconnect compound.

(25) FIG. **3***g* illustrates SIP module **138** with encapsulant **136** deposited over electrical components **130***a*-**130***c* and interconnect substrate **120**. Electrical components **140***a*-**140***b* are electrically and mechanically connected to conductive layers **122** and electrical interconnect compound **144** is deposited on the exposed vertical interconnect vias **124** of substrate **120**. FIG. **3***h* is a side view of SIP module **138**, from FIG. **3***g*, showing insulating layers **123** and electrical components **140***a*-**140***b* electrically and mechanically connected to conductive layers **122** and electrical interconnect compound **144** deposited on the exposed vertical interconnect vias **124** on side surface **141** of substrate **120**.

(26) To perform the steps of FIGS. **2***a*-**2***h* and **3***a*-**3***h*, a plurality of SIP modules **138** are mounted to carrier **160** with a vertical orientation, as shown in FIG. **4***a*. Carrier **160** can be a film with an adhesive layer. SIP modules **138** adhere to the adhesive layer of carrier **160**. Alternatively, carrier **162** can be a rigid support, such as metal or plastic with a plurality of slots **164** sized to

accommodate SIP modules **138**, as shown in FIG. **4***b*. The plurality of SIP modules **138** are positioned within slots **164** with a vertical orientation.

(27) In FIG. **4***c*, electrical interconnect compound **144**, such as solder or electrically conductive paste, is deposited over the exposed vertical interconnect vias **124** on side surface **141** of SIP modules **138** on carrier **160**. In particular, electrical interconnect compound **144** is deposited using nozzle **166**, as shown in FIG. **4***d*. Nozzle **166** is tilted at an angle θ of about 25 degrees to reliably and accurately deliver electrical interconnect compound **144** to a smaller area on vertical interconnect vias **124**. Tilt dispensing with nozzle **166** enables precise dispensing of electrical interconnect compound **144** on the pad of the vertically-oriented substrate. Nozzle **166** tilted at an angle is able to move to a target location for precision delivery. Accordingly, nozzle **166** deposits electrical interconnect compound **144** on a pad of vertical interconnect vias **124**. Alternatively, electrical interconnect compound **144** can be deposited by paste printing or other paste dispensing technology.

(28) In FIG. **4***e*, electrical components **140***a*-**140***b* are positioned over electrical interconnect compound **144** on side surface **141** using pick and place tool **168**. In FIG. **4***f*, electrical components **140***a*-**140***b* are electrically and mechanically connected to conductive layers **122** and electrical interconnect compound **144** deposited on the exposed vertical interconnect vias **124** on side surface **141** of substrate **120** using laser **170**. The laser assist bonding with laser **170** enables component mounting and avoids package warpage as compared to other mass reflow. An underfill material **172**, such as an epoxy resin, can be deposited between electrical components **140***a*-**140***b* and interconnect substrate **120**, as shown in FIG. **4***g*.

(29) In another embodiment, SIP module **180** may have electrical components **174**, similar to electrical components **140***a*-**140***b*, disposed on one, two, three, or four side surfaces **141**. FIG. **5***a* shows electrical components **174** electrically and mechanically connected to conductive layers **122** and electrical interconnect compound **176** deposited on the exposed vertical interconnect vias **124** on multiple side surface **141** of substrate **120**. FIG. **5***b* is a top view of SIP module **180** showing electrical components **174** electrically and mechanically connected to conductive layers **122** and electrical interconnect compound **176** deposited on the exposed vertical interconnect vias **124** on all four side surfaces **141** of substrate **120**. An underfill material **182**, such as an epoxy resin, can be deposited between electrical components **174** and interconnect substrate **120**.

(30) SIP module **138** or **180** with electrical components **140***a*-**140***b* or **174** disposed on one or more side surfaces **141** of interconnect substrate **120**, as see in FIGS. **2***f*, **3***h*, **4***f*, and **5***b*, decreases the size of the package and increases space utilization.

(31) Electrical components **130***a*-**130***c* may contain IPDs that are susceptible to or generate EMI, RFI, harmonic distortion, and inter-device interference. For example, the IPDs contained within electrical components **130***a*-**130***c* provide the electrical characteristics needed for high-frequency applications, such as resonators, high-pass filters, low-pass filters, band-pass filters, symmetric Hi-Q resonant transformers, and tuning capacitors. In another embodiment, electrical components **130***a*-**130***b* contain digital circuits switching at a high frequency, which could interfere with the operation of IPDs in the SIP module.

(32) SIP module **138** includes high speed digital and RF electrical components **130***a*-**130***c*, highly integrated for small size and low height, and operating at high clock frequencies. In FIG. **6**, electromagnetic shielding layer **190** is formed or disposed over encapsulant **136** and a portion of interconnect substrate **120** by conformal application of shielding material. Shielding layer **190** can be one or more layers of Al, Cu, Sn, Ni, Au, Ag, or other suitable conductive material. Alternatively, shielding layer **190** can be carbonyl iron, stainless steel, nickel silver, low-carbon steel, silicon-iron steel, foil, conductive resin, carbon-black, aluminum flake, and other metals and composites capable of reducing or inhibiting the effects of EMI, RFI, and other inter-device interference. In addition, shielding layer **190** covers side surfaces of encapsulant **136**, as well as one or more side surfaces of substrate **120**. Electromagnetic shielding layer **190** reduces or inhibits

EMI, RFI, and other inter-device interference, for example as radiated by high-speed digital devices, from affecting neighboring devices within or adjacent to SIP module **138**.

(33) FIG. **7** illustrates electronic device **300** having a chip carrier substrate or PCB **302** with a plurality of semiconductor packages mounted on a surface of PCB **302**, including SIP modules **138** and **180**. Electronic device **300** can have one type of semiconductor package, or multiple types of semiconductor packages, depending on the application.

(34) Electronic device **300** can be a stand-alone system that uses the semiconductor packages to perform one or more electrical functions. Alternatively, electronic device **300** can be a subcomponent of a larger system. For example, electronic device **300** can be part of a tablet, cellular phone, digital camera, communication system, or other electronic device. Alternatively, electronic device **300** can be a graphics card, network interface card, or other signal processing card that can be inserted into a computer. The semiconductor package can include microprocessors, memories, ASIC, logic circuits, analog circuits, RF circuits, discrete devices, or other semiconductor die or electrical components. Miniaturization and weight reduction are essential for the products to be accepted by the market. The distance between semiconductor devices may be decreased to achieve higher density.

(35) In FIG. **7**, PCB **302** provides a general substrate for structural support and electrical interconnect of the semiconductor packages mounted on the PCB. Conductive signal traces **304** are formed over a surface or within layers of PCB **302** using evaporation, electrolytic plating, electroless plating, screen printing, or other suitable metal deposition process. Signal traces **304** provide for electrical communication between each of the semiconductor packages, mounted components, and other external system components. Traces **304** also provide power and ground connections to each of the semiconductor packages.

(36) In some embodiments, a semiconductor device has two packaging levels. First level packaging is a technique for mechanically and electrically attaching the semiconductor die to an intermediate substrate. Second level packaging involves mechanically and electrically attaching the intermediate substrate to the PCB. In other embodiments, a semiconductor device may only have the first level packaging where the die is mechanically and electrically mounted directly to the PCB. For the purpose of illustration, several types of first level packaging, including bond wire package **306** and flipchip **308**, are shown on PCB **302**. Additionally, several types of second level packaging, including ball grid array (BGA) **310**, bump chip carrier (BCC) **312**, land grid array (LGA) **316**, multi-chip module (MCM) or SIP module **318**, quad flat non-leaded package (QFN) **320**, quad flat package **322**, embedded wafer level ball grid array (eWLB) **324**, and wafer level chip scale package (WLCSP) **326** are shown mounted on PCB **302**. In one embodiment, eWLB **324** is a fan-out wafer level package (Fo-WLP) and WLCSP **326** is a fan-in wafer level package (Fi-WLP). Depending upon the system requirements, any combination of semiconductor packages, configured with any combination of first and second level packaging styles, as well as other electronic components, can be connected to PCB **302**. In some embodiments, electronic device **300** includes a single attached semiconductor package, while other embodiments call for multiple interconnected packages. By combining one or more semiconductor packages over a single substrate, manufacturers can incorporate pre-made components into electronic devices and systems. Because the semiconductor packages include sophisticated functionality, electronic devices can be manufactured using less expensive components and a streamlined manufacturing process. The resulting devices are less likely to fail and less expensive to manufacture resulting in a lower cost for consumers.

(37) While one or more embodiments of the present invention have been illustrated in detail, the skilled artisan will appreciate that modifications and adaptations to those embodiments may be made without departing from the scope of the present invention as set forth in the following claims.

Claims

1. A semiconductor device, comprising: an interconnect substrate including a conductive layer internal to the interconnect substrate, wherein the conductive layer includes a first conductive via exposed from a side surface of the interconnect substrate; a first electrical component disposed over a major surface of the interconnect substrate; and a second electrical component disposed in contact with the first conductive via on the side surface of the interconnect substrate.
2. The semiconductor device of claim 1, further including an electrical interconnect compound disposed over the first conductive via exposed from the side surface of the interconnect substrate.
3. The semiconductor device of claim 1, further including a plurality of second electrical components disposed on two or more side surfaces of the interconnect substrate.
4. The semiconductor device of claim 1, wherein the first conductive via includes a plurality of stacked conductive vias and the second electrical component is disposed in contact with the stacked conductive vias.
5. The semiconductor device of claim 1, further including an encapsulant deposited over the first electrical component and interconnect substrate.
6. The semiconductor device of claim 5, further including a shielding layer formed over the encapsulant.
7. The semiconductor device of claim 1, wherein the first conductive via extends from a first surface of the interconnect substrate to a second surface of the interconnect substrate opposite the first surface of the interconnect substrate.
8. The semiconductor device of claim 1, wherein the conductive layer further includes a second conductive via vertically and laterally offset from the first conductive via.
9. A semiconductor device, comprising: a substrate including a first conductive via exposed from a side surface of the substrate and a second conductive via exposed from the side surface of the substrate; a first electrical component disposed over a major surface of the substrate; and a second electrical component disposed over the first conductive via and second conductive via on the side surface of the substrate.
10. The semiconductor device of claim 9, further including an electrical interconnect compound disposed over the side surface of the substrate.
11. The semiconductor device of claim 9, further including a plurality of second electrical components disposed over two or more side surfaces of the substrate.
12. The semiconductor device of claim 9, wherein providing the substrate includes: forming a conductive layer internal to the substrate, wherein the the first conductive via is part of the conductive layer; and disposing the second electrical component is disposed over the first conductive via.
13. The semiconductor device of claim 9, wherein the substrate includes a plurality of stacked conductive vias and the second electrical component is disposed over the stacked conductive vias.
14. The semiconductor device of claim 9, further including an encapsulant deposited over the first electrical component and substrate.
15. The semiconductor device of claim 14, further including a shielding layer formed over the encapsulant.
16. The semiconductor device of claim 9, wherein the first conductive via extends from a first surface of the substrate to a second surface of the substrate opposite the first surface of the substrate.
17. The semiconductor device of claim 9, wherein the substrate includes a second conductive via vertically and laterally offset from the first conductive via.
18. A method of making a semiconductor device, comprising: providing an interconnect substrate including a conductive layer internal to the interconnect substrate, wherein the conductive layer

includes a first conductive via exposed from a side surface of the interconnect substrate; disposing a first electrical component over a major surface of the interconnect substrate; and disposing a second electrical component over the first conductive via exposed from a side surface of the interconnect substrate.

19. The method of claim 18, further including disposing an electrical interconnect compound over the first conductive via exposed from the side surface of the interconnect substrate.

20. The method of claim 18, further including disposing a plurality of second electrical components over two or more side surfaces of the interconnect substrate.

21. The method of claim 18, wherein providing the interconnect substrate includes: forming a plurality of stacked conductive vias through the interconnect substrate; and disposing the second electrical component over the stacked conductive vias.

22. The method of claim 18, further including depositing an encapsulant over the first electrical component and interconnect substrate.

23. The method of claim 22, further including forming a shielding layer over the encapsulant.

24. The method of claim 18, wherein the first conductive via extends from a first surface of the interconnect substrate to a second surface of the interconnect substrate opposite the first surface of the interconnect substrate.

25. The method of claim 18, wherein the conductive layer further includes a second conductive via vertically and laterally offset from the first conductive via.

26. A method of making a semiconductor device, comprising: providing a substrate including a first conductive via exposed from a side surface of the substrate and a second conductive via exposed from the side surface of the substrate; disposing a first electrical component over a major surface of the substrate; and disposing a second electrical component over the first conductive via and second conductive via on the side surface of the substrate.

27. The method of claim 26, further including disposing an electrical interconnect compound over the side surface of the substrate.

28. The method of claim 26, further including disposing a plurality of second electrical components over two or more side surfaces of the substrate.

29. The method of claim 26, further including depositing an encapsulant over the first electrical component and substrate.

30. The method of claim 29, further including forming a shielding layer over the encapsulant.

31. The method of claim 26, wherein the first conductive via extends from a first surface of the substrate to a second surface of the substrate opposite the first surface of the substrate.

32. The method of claim 26, wherein the substrate includes a second conductive via vertically and laterally offset from the first conductive via.